

Okidi Norbert

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PROFILE SUMMARY

Computer Science student at Uganda Christian University with a strong foundation in embedded systems, firmware development, and full-stack software development. Highly proficient in AI-driven engineering, actively leveraging advanced AI agents (Cursor, Antigravity, Claude, ChatGPT) across the entire software development lifecycle—from architecture and documentation to scripting, deep debugging, and deployment. Deeply passionate about blurring the lines between AI, hardware hacking, and offensive security. Also a competitive semi-professional basketball player, bringing discipline and strategic thinking to engineering challenges.

EDUCATION

Uganda Christian University, Mukono
Bachelor of Science in Computer Science

Expected Graduation: July 2026
Current CGPA: 4.39/5.0

- **Key Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Database Programming, Web Programming, Cyber Security & Digital Forensics, Data Science, Embedded Systems & Microcontroller Programming, Robotics & Control Systems, Very Large Scale Integration (VLSI), Digital Electronics & Logic Design.

Buddo Secondary School — Uganda Advanced Certificate of Education (UACE)
St. Joseph's College Layibi — Uganda Certificate of Education (UCE)

2022
2019

CERTIFICATIONS

- **Cybersecurity Job Simulation** – Commonwealth Bank
- **Digital Forensics & Incident Response (DFIR)**
- **Cloud Foundations**

TECHNICAL SKILLS

- **Programming Languages:** C, C++, Python, Java, TypeScript, Bash, JavaScript, PHP
- **Embedded Systems & Hardware:** Microcontrollers, Firmware Debugging, Raspberry Pi, Arduino, UART, Serial Communication, PLC Fundamentals, Industrial Automation
- **AI-Assisted Development:** Master of AI-driven engineering workflows (Cursor, Antigravity, Copilot, Lovable, Claude, ChatGPT) for rapid prototyping, complex debugging, and end-to-end SDLC automation.
- **AI & Machine Learning:** PyTorch, TensorFlow, OpenCV, Scikit-learn, YOLO
- **Web & Databases:** React, Next.js, Django, Flask, HTML/CSS, Tailwind CSS, PostgreSQL, MySQL, MongoDB, Supabase
- **DevOps & Deployment:** Vercel, Hugging Face, AWS, Docker, Git, CI/CD Pipelines
- **Cybersecurity & OS:** Linux (Kali, Debian, Ubuntu), Windows, Ethical Hacking, Penetration Testing, Digital Forensics
- **Creative Tools:** Adobe Photoshop, DaVinci Resolve, CapCut, Canva, Motion Graphics

PROFESSIONAL EXPERIENCE

Digital Transformation & Growth Manager — Alora Initiative

Oct 2025 – Mar 2026

- Led digital transformation initiatives to drive growth and streamline organizational processes.

IT Intern — Uganda Industrial Research Institute (UIRI)

May 2025 – July 2025

- Gained hands-on exposure to embedded systems and firmware development.
- Worked with PLC-based industrial automation systems and assisted in hardware-software integration.
- Developed and tested C/C++ hardware control programs and validation scripts.
- Participated in an exhibition showcasing a custom-built robotic arm project.

Independent Technical Consultant & Hardware Engineer

2021 – Present

- Founded and operated a technical consultancy service resolving hardware and software issues for 150+ clients within the university community.
- Diagnosed low-level hardware faults, performed data recovery, configured operating systems (macOS, Linux, Windows), and executed component-level hardware upgrades.
- Automated system diagnostics and repetitive maintenance tasks using custom Python and Bash scripting.
- Configured local network devices, troubleshooted routing issues, and performed deep-system software debugging.

Independent Content Creator & Video Editor

2021 – Present

- Created and edited high-engagement short-form and long-form video content for social media platforms.
- Produced cinematic basketball edits, highlights, and promotional graphics with advanced transitions.
- Utilized AI-assisted editing workflows, sound synchronization, and digital storytelling techniques.

PROJECTS

BAKO-AI — Sports Analytics & Computer Vision Platform 2026

- Engineered an advanced real-time computer vision engine using PyTorch, OpenCV, and custom YOLO architectures for automated basketball tracking (player, ball, and court keypoints).
- Architected a 4-phase shot-form biometric skill diagnostics pipeline and a 7-module tactical analytics suite.
- Developed a robust full-stack system featuring automated video ingestion, Supabase/PostgreSQL database management, and interactive React/Next.js analytics dashboards.

CineQuest — AI Movie Discovery Platform 2025

- Spearheaded full-stack integration and recommendation model engineering for an interactive media search platform.
- Leveraged state-of-the-art AI agent workflows (Cursor, Antigravity) to accelerate API routing, CI/CD deployment, and high-performance scripting.

Embedded Robotic Arm Control System 2025

- Designed and fabricated a multi-degree-of-freedom (DOF) robotic arm prototype controlled via custom C/C++ firmware.
- Implemented low-level actuator control utilizing PWM-driven servo systems, UART serial protocols, and real-time sensor integration.

SPORTS & EXTRACURRICULAR

UCU Canons – National Basketball League (NBL) Shooting Guard, Small Forward, Power Forward

- Represented Uganda Christian University at the 2024 World University 3x3 Championship in Xiamen, China.
- Demonstrated high-level teamwork, leadership, and performance under pressure with 6+ years of competitive play.

Other Interests: 3D Design & Printing (Designing STL files), Computer & Electronics Repair, Cybersecurity Research.